UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

SEPTEMBER EXAMINATION

UCCN2243 INTERNETWORKING PRINCIPLES AND PRACTICES

THURSDAY, 12 OCTOBER 2023 TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF COMPUTER SCIENCE (HONOURS)

Instruction to Candidates :

This question paper consists of THREE (3) questions in Section A and TWO (2) questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than ONE (1) question in section B, marks will only be awarded for the FIRST question in that section in the order the candidate submits the answers.

Candidates are allowed to use calculator.

Answer questions only in the answer booklet provided.

Section A (Compulsory Questions)

Q1. You are assigned the following network 202.1.144.0/20. Using variable-length subnet masking (VLSM), create subnets as required for the network shown in Figure 1 (assume IP subnet-zero). Ensure that you always utilize the next available network in your IP plan.

(a) **Calculate** the network address, and subnet masks for all the networks (Area 1 to Area 9). Hint: The number of usable addresses in Area 4 to Area 8 corresponds to the number of gateway interfaces in their respective subnets. (15 marks)

(b) **Determine** the gateway address of Area 3 and Area 9. Assume the respective gateway address is the second-last usable address. (4 marks)

(c) **Determine** the summarized route of Area 1 to Area 9, with the longest possible network prefix. (6 marks)

[Total : 25 marks]

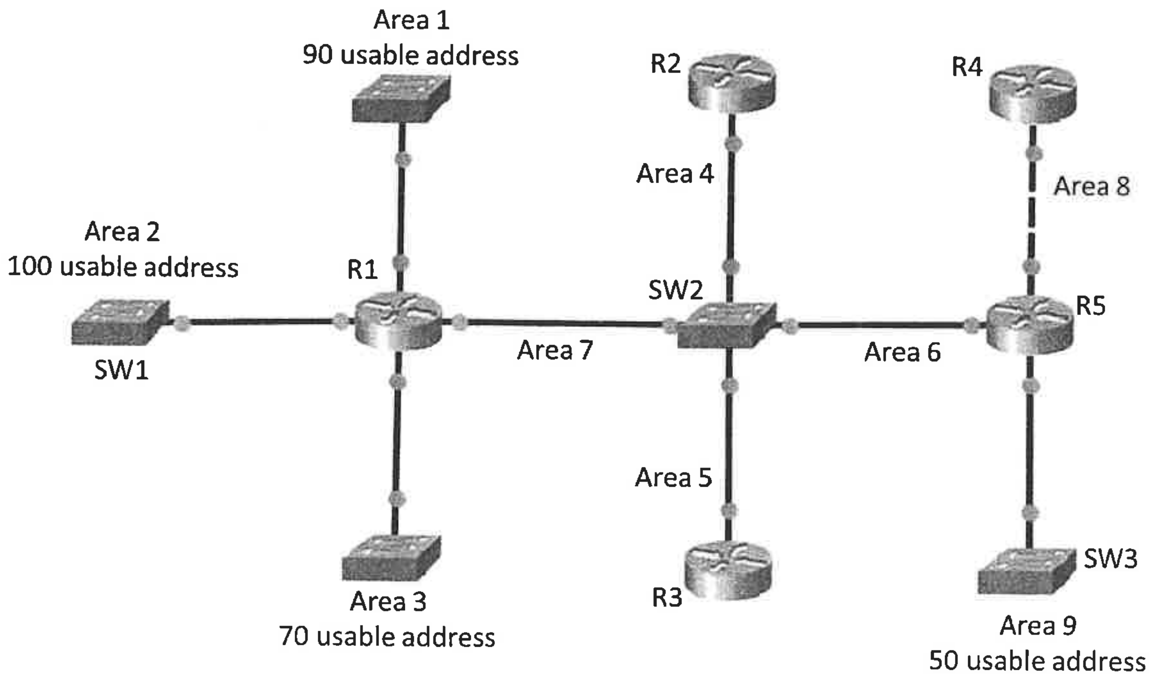

Figure 1

Q2. Refer to the network in Figure 2. Company AA has a public IP address of 200.0.0.1. Network Address Translation (NAT) is configured on routers R1 and R2 to enable access of Company AA web servers from the outside network. Assume that the necessary IP addresses have been configured in all the router ports.

(a) Due to security reasons, R1 will only translate hosts with the IP address 172.16.1.2 to the public IP for Internet access. **Show** the relevant ACL and Port Address Translation configurations on routers R1 and R2. (Hint: You will only need to utilize the information given in Figure 2 for your setup).
(15 marks)

(b) **Show** the relevant Network Address Translation (NAT) configurations of the routers (R1 and R2) to enable Outside PC access to Web Server 1 and Web Server 2 using the respective HTTP address. (Hint: You will only need to utilize the information given in Figure 2 for your setup). (8 marks)

(c) **Show** the IP routes configurations of routers R1 and R2 if suppose they utilize default IP routes. (Hint: You will only need to utilize the information given in Figure 2 for your setup). (2 marks)
[Total : 25 marks]

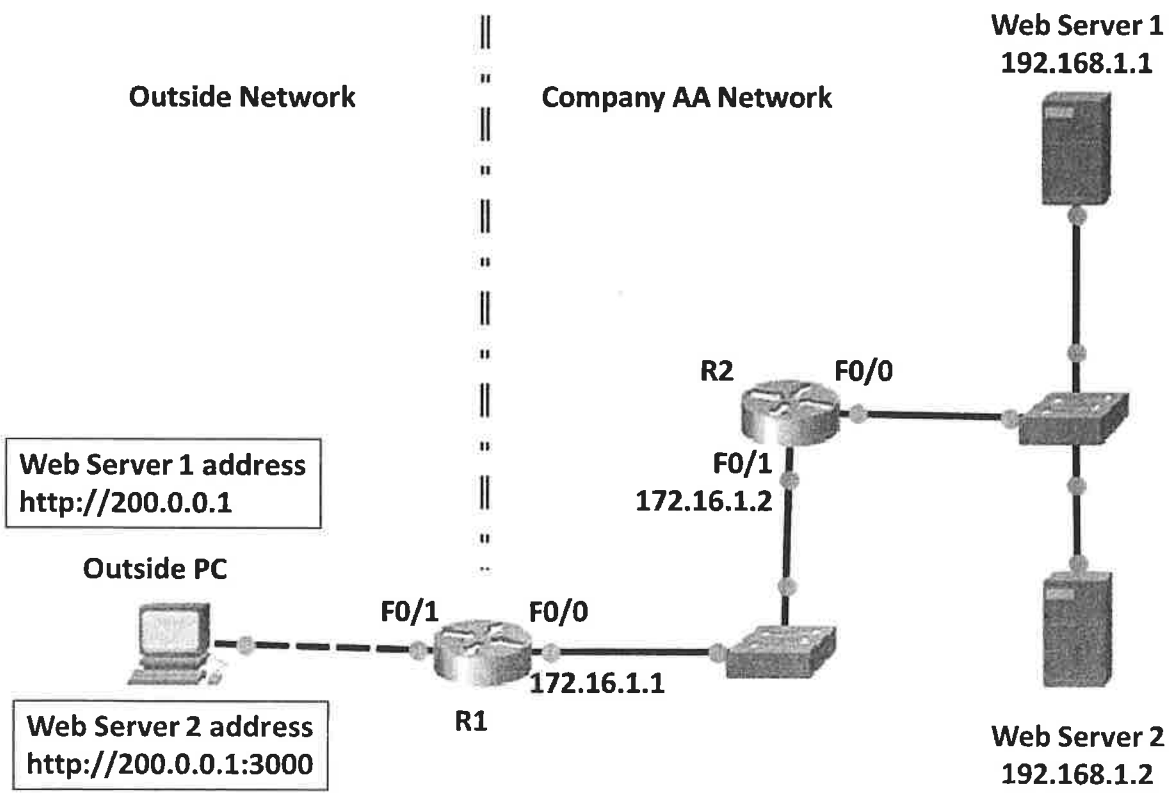

Figure 2

Q3. Figure 3 shows two IPv6 only network (Subnet 1 and Subnet2) connected over an existing IPv4 only network.

(a) **Discuss** THREE (3) types of techniques to enable Subnet 1 to communicate with Subnet 2 over the existing IPv4 only infrastructure. (6 marks)

(b) Given the IPv6 address for Server 2 is 2001::AABB:CCDD:000A/101. **Show** all the necessary commands to configure Router 2 with the 3rd usable IP address as the gateway for Subnet 2. (5 marks)

(c) Continue from Q3.(b), **determine** the IPv6 address of Server 1, assuming that it is assigned with the last usable address of the subnet. (3 marks)

(d) **Determine** the IPv6 address and link-local address of PC2 using eui-64, if its media access control (MAC) address is 00-11-83-5D-77-8A. (7 marks)

(e) **Show** the necessary commands to configure interface G1 of Router 1 and interface G1 of Router 2 with dynamic routing using the IPv6 RIP scheme: IPP2243. You may assume the basic network setup on the routers has been done. (4 marks)

[Total : 25 marks]

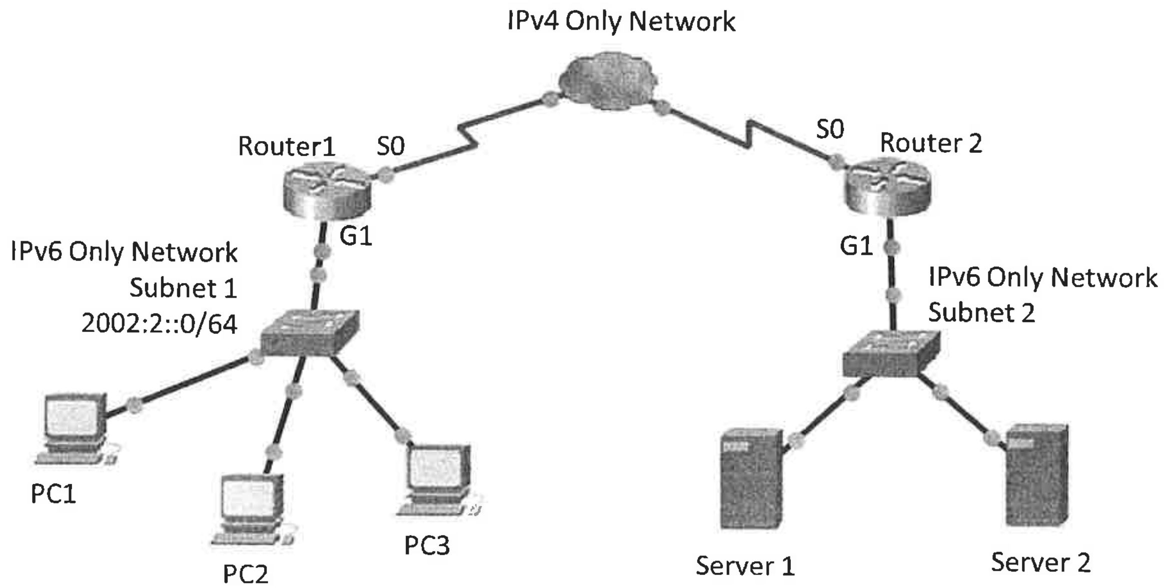

Figure 3

Section B (Choose Any One Question)

Q4. Given the network shown in Figure 4.

(a) Table 1 shows the initial ARP (Address Resolution Protocol) table of PC1. **Redraw** the table and **specify** the corresponding MAC address. (4 marks)

Table 1

IP address	MAC address
192.168.1.2	
192.168.1.254	
192.168.2.1	
200.1.1.200	

(b) Supposed PC2 launches an ARP spoofing attack on PC1. **Redraw** Table 1 and **specify** the corresponding MAC address after the ARP spoofing attack. (4 marks)

(c) **Redraw** Figure 4 to illustrate the packet flow between PC1 and Server 1 after the ARP spoofing attack by PC2. (6 marks)

(d) Using iptables, **configure** PC1 to drop any packets from PC2 (4 marks)

(e) Continue from Q4.(d), configure PC1 to only accept ping requests that is slower than 2 per second (3 marks)

(f) Supposed router R1 is installed with iptables. Configure R1 to reject packets to the destination IP 200.10.10.11. (4 marks)

[Total : 25 marks]

Q4. (Continued)

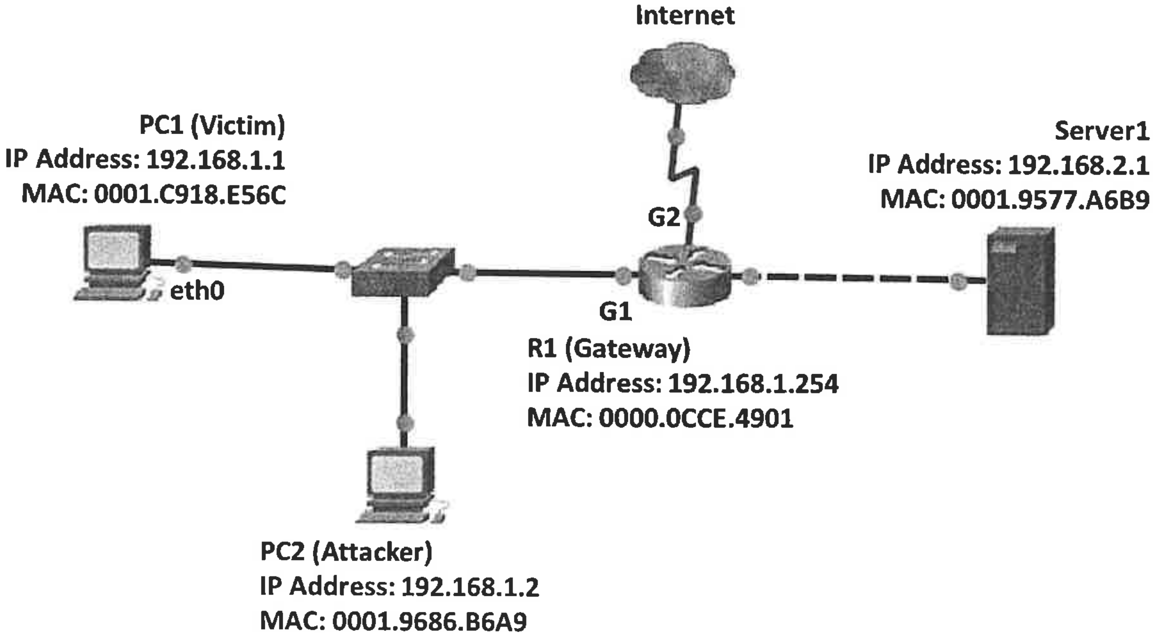

Figure 4

Q5. (a) Figure 5 shows the data dump (in hexadecimal) at the beginning of an IP packet. Suppose the datagram will be routed to an Ethernet network, answer the following:

(i) **Calculate** the size (in bytes) of the initial IP data before fragmentation. (4 marks)

(ii) **Determine** the value of the header options. (2 marks)

(iii) **Calculate** the total length for each of the fragments (assume similar options value). (8 marks)

(iv) **Determine** the fragment offset for each of the fragments. (3 marks)

Q5. (Continued)

46 00 0F A0 10 00 00 00 ....

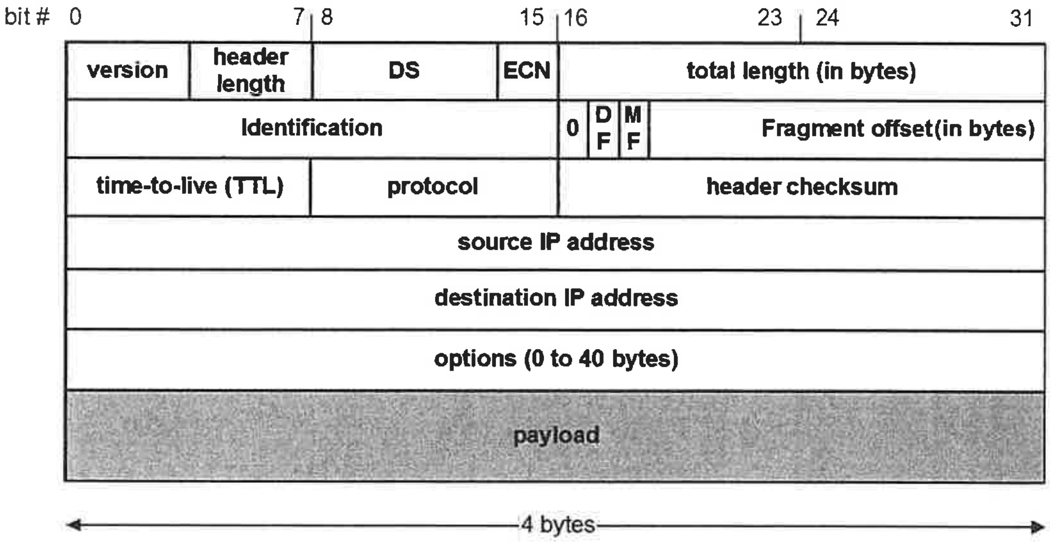

Figure 5

(b) Connection-oriented service is among the options provided by the Transport layer. **Describe** FOUR (4) characteristics of this connection. (8 marks)

[Total : 25 marks]
